Taylor and Maclaurin Series

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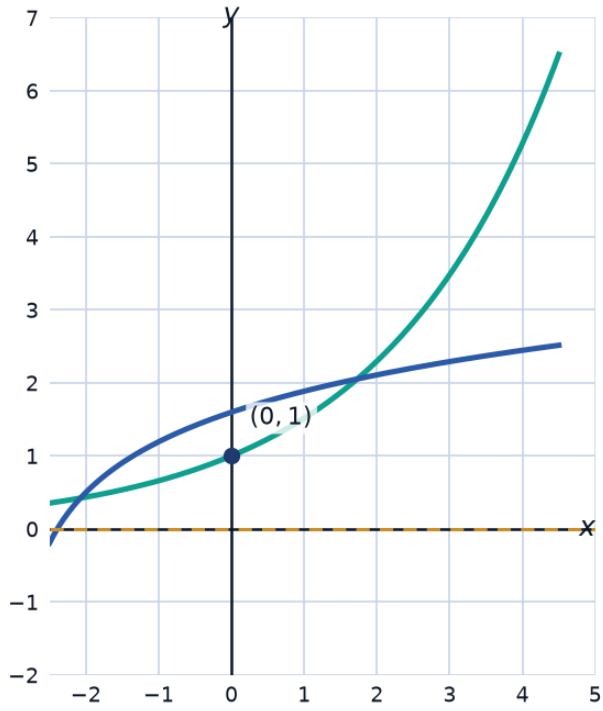
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: The Taylor series formula

State the general Taylor series formula for $f(x)$ centered at $x = a$.

2 [Foundational] • Target: The Taylor series formula

Find the Taylor series for $f(x) = \ln x$ centered at $a = 1$, up through the $(x - 1)^3$ term.



3 [Foundational] • Target: Common Maclaurin series

State the Maclaurin series for e^x, $\sin x$, and $\cos x$.

4 [Foundational] • Target: Common Maclaurin series

Use the Maclaurin series for e^x to find the Maclaurin series for e^{-x^2}.

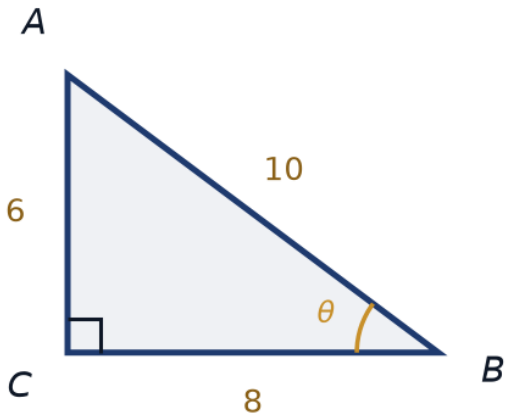
5 [Foundational] • Target: Using known series to build new series

Use the Maclaurin series for $\sin x$ to find the first three nonzero terms of the Maclaurin series for $x \sin x$.

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6 [Foundational] • Target: Using known series to build new series

Use the known Taylor series for $\cos x$ to estimate $\cos(0.1)$ using the first three nonzero terms.



7 [Foundational] • Target: The Lagrange error bound

State the Lagrange Error Bound formula for the remainder $R_n(x)$ of a Taylor polynomial of degree n.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The Lagrange error bound

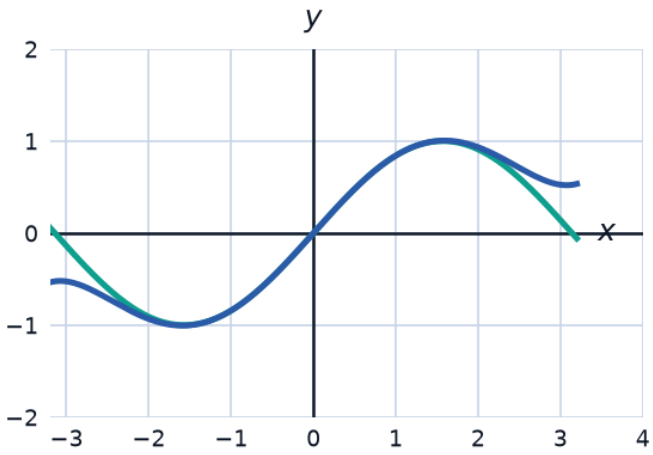
Use the Lagrange Error Bound to estimate the maximum error when approximating $\sin(0.5)$ with the Taylor polynomial $P_3(x) = x - \frac{x^3}{6}$ (degree 3, centered at 0), given $|f^{(4)}(z)| \le 1$ for all z.

9 [Intermediate] • Target: Radius of convergence of Taylor series

State the radius of convergence of the Maclaurin series for e^x, $\sin x$, and $\cos x$.

10 [Intermediate] • Target: Visualizing Taylor polynomial approximations

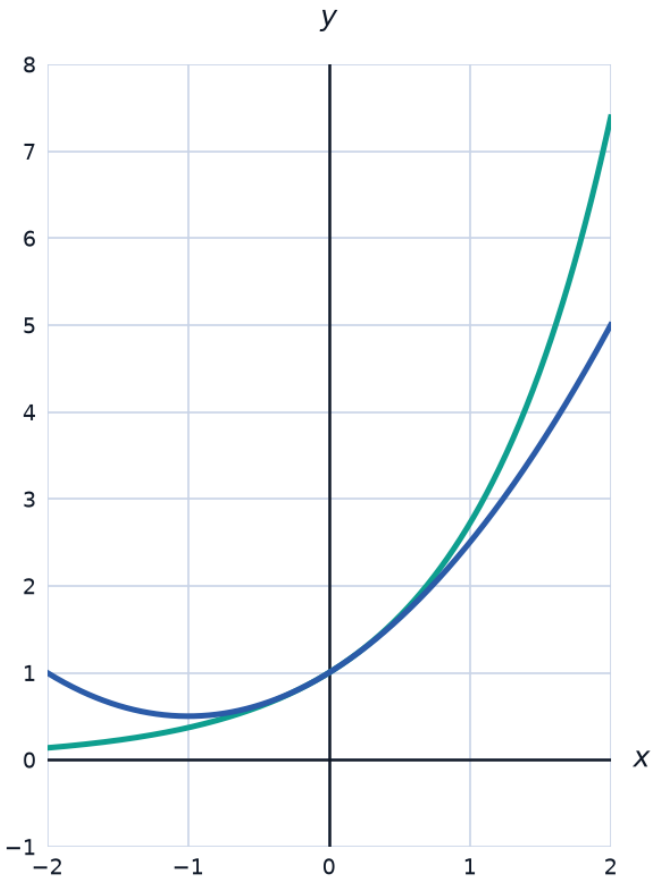
Compare $f(x) = \sin x$ to its degree-5 Maclaurin polynomial $P_5(x) = x - \frac{x^3}{6} + \frac{x^5}{120}$ on $[-\pi, \pi]$, and describe where the approximation is best and worst.



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11 [Intermediate] • Target: Visualizing Taylor polynomial approximations

Compare $f(x) = e^x$ to its degree-2 Maclaurin polynomial $P_2(x) = 1 + x + \frac{x^2}{2}$ on $[-2, 2]$, and identify visually where P_2 underestimates or overestimates e^x.



12 [Intermediate] • Target: Building Taylor series at a nonzero center

Find the Taylor series for $f(x) = \cos x$ centered at $a = \frac{\pi}{2}$, up through the cubic term.

13 [Intermediate] • Target: Applying the Lagrange Error Bound to bound accuracy requirements

Using the degree-4 Maclaurin polynomial for $\cos x$ and the bound $|f^{(5)}(z)| \le 1$, find the maximum error in approximating $\cos(0.3)$.

14 [Intermediate] • Target: The binomial series

State the binomial series for $(1 + x)^k$ up through the quadratic term, and use it to find the Maclaurin series for $\sqrt{1 + x}$ (i.e. $k = \frac{1}{2}$) up through the quadratic term.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

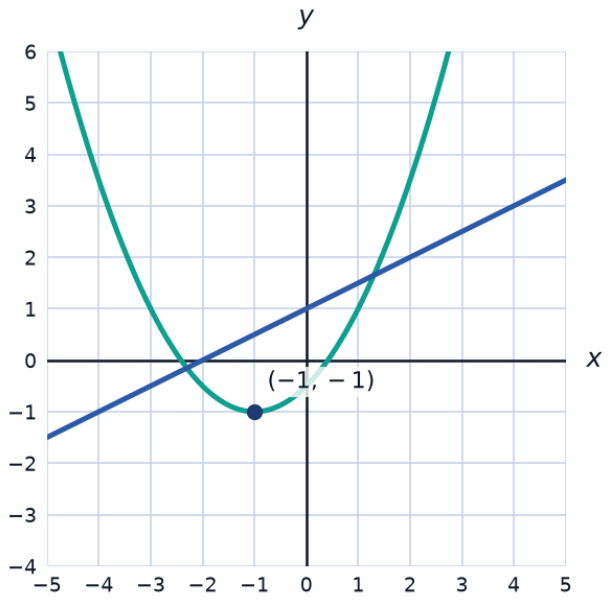
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15 [Challenging] • Target: The binomial series • Cross-Lesson: 3.02 Vector-Valued Functions

Use the series found above to approximate $\sqrt{1.1}$ using the first three terms.

16 [Challenging] • Target: Deriving new Maclaurin series from known ones • Cross-Lesson: 4.01 Polar Curves and Derivatives

Use the known series for e^x to find the Maclaurin series for $\cosh x = \frac{e^x + e^{-x}}{2}$, up through the quartic term.



17 [Challenging] • Target: Using known series to build new series • Cross-Lesson: 4.02 Area in Polar Coordinates

For $x = t^3,\ y = t^2$, find $\frac{dy}{dx}$ at $t = 1$.

18 [Challenging] • Target: The Lagrange error bound • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

For $x = t^2,\ y = 2t$, find $\frac{dy}{dx}$.

19 [Challenging] • Target: Radius of convergence of Taylor series • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Using $\sin(3x) = \sum_{n=0}^{\infty} \frac{(-1)^n (3x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^5.

20 [Challenging] • Target: Visualizing Taylor polynomial approximations • Cross-Lesson: 6.01 Power Series and Interval of Convergence

Using $\sin(3x) = \sum_{n=0}^{\infty} \frac{(-1)^n (3x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] Target: The Taylor series formula

Using $e^{3x} = \sum_{n=0}^{\infty} \frac{(3x)^n}{n!}$, find the coefficient of x^5.

a. $\frac{121}{40}$

b. $\frac{81}{8}$

c. $\frac{1}{40}$

d. $\frac{81}{40}$

22 [Foundational] Target: Common Maclaurin series

Using $e^{6x} = \sum_{n=0}^{\infty} \frac{(6x)^n}{n!}$, find the coefficient of x^2.

a. 18

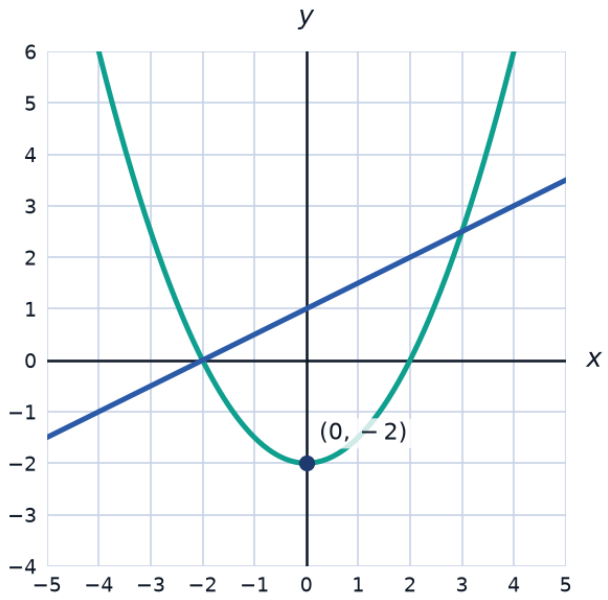
b. 19

c. 3

d. 36

23 [Foundational] Target: Using known series to build new series

Using $e^{8x} = \sum_{n=0}^{\infty} \frac{(8x)^n}{n!}$, find the coefficient of x^4.



a. $\frac{512}{3}$

b. $\frac{2048}{3}$

c. $\frac{515}{3}$

d. $\frac{1}{3}$

24 [Foundational] Target: The Lagrange error bound

Using $e^{8x} = \sum_{n=0}^{\infty} \frac{(8x)^n}{n!}$, find the coefficient of x^3.

a. $\frac{4}{3}$

b. 256

c. $\frac{259}{3}$

d. $\frac{256}{3}$

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25 [Foundational] Target: Radius of convergence of Taylor series

Using $e^{2x} = \sum_{n=0}^{\infty} \frac{(2x)^n}{n!}$, find the coefficient of x^2.

a. 4

b. 3

c. 2

d. 1

26 [Foundational] Target: Visualizing Taylor polynomial approximations

Using $e^{6x} = \sum_{n=0}^{\infty} \frac{(6x)^n}{n!}$, find the coefficient of x^5.

a. 324

b. $\frac{324}{5}$

c. $\frac{329}{5}$

d. $\frac{1}{20}$

27 [Foundational] Target: Building Taylor series at a nonzero center

Using $e^{8x} = \sum_{n=0}^{\infty} \frac{(8x)^n}{n!}$, find the coefficient of x^2.

a. 32

b. 64

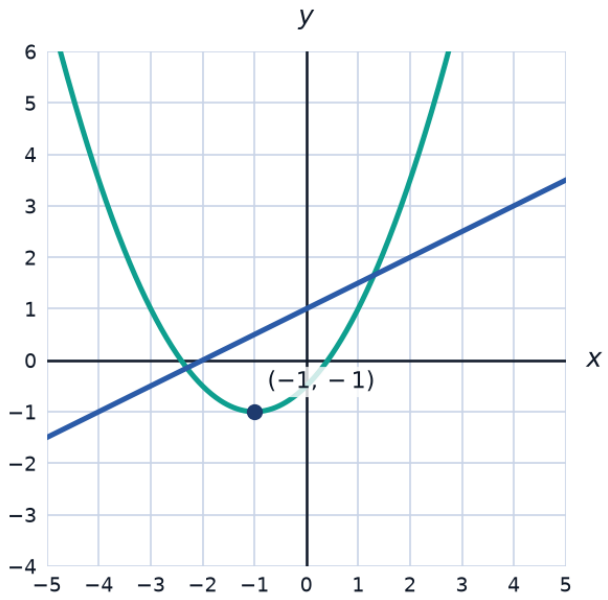
c. 33

d. 4

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28 [Foundational] • Target: Applying the Lagrange Error Bound to bound accuracy requirements

Using $\sin(6x) = \sum_{n=0}^{\infty} \frac{(-1)^n (6x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3.



a. -35

b. 36

c. -36

d. 108

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The binomial series

Using $\sin(5x) = \sum_{n=0}^{\infty} \frac{(-1)^n (5x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3.

a. $\frac{125}{2}$

b. $-\frac{125}{6}$

c. $\frac{125}{6}$

d. $-\frac{119}{6}$

30 [Intermediate] • Target: Deriving new Maclaurin series from known ones

Using $\sin(4x) = \sum_{n=0}^{\infty} \frac{(-1)^n (4x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3.

a. $-\frac{29}{3}$

b. $\frac{32}{3}$

c. 32

d. $-\frac{32}{3}$

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31 [Intermediate] • Target: Common Maclaurin Series ($e^x, \sin x, \cos x$)

Using $\sin(2x) = \sum_{n=0}^{\infty} \frac{(-1)^n (2x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3.

a. 4

b. $-\frac{1}{3}$

c. $\frac{4}{3}$

d. $-\frac{4}{3}$

32 [Intermediate] • Target: Constructing Taylor Polynomials from Derivatives

Using $\sin(9x) = \sum_{n=0}^{\infty} \frac{(-1)^n (9x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3.

a. $-\frac{241}{2}$

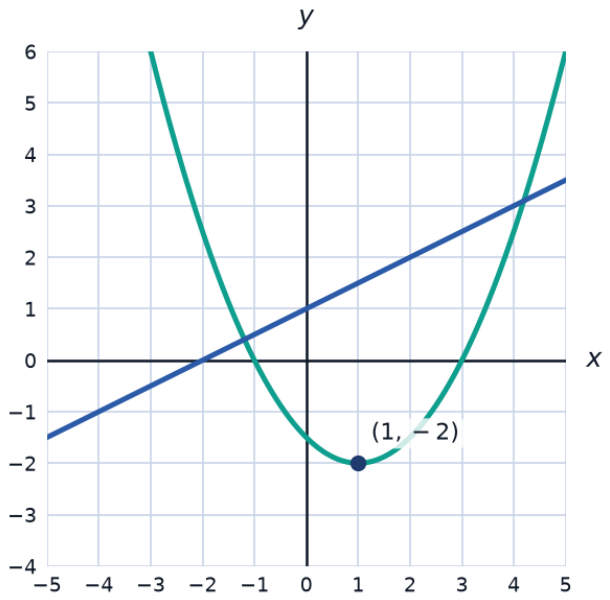
b. $-\frac{243}{2}$

c. $\frac{729}{2}$

d. $\frac{243}{2}$

33 [Intermediate] • Target: Manipulating Known Taylor Series

Using $\sin(5x) = \sum_{n=0}^{\infty} \frac{(-1)^n (5x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^5.



a. $-\frac{625}{24}$

b. $\frac{649}{24}$

c. $\frac{625}{24}$

d. $\frac{3125}{24}$

34 [Intermediate] • Target: Lagrange Error Bound (Taylor's Remainder Theorem)

Use the Maclaurin series for e^u (with $u = 3x^2$) to find the coefficient of x^8 in the Maclaurin series for e^{3x^2}.

a. $\frac{2187}{8}$

b. $\frac{9}{4480}$

c. $\frac{35}{8}$

d. $\frac{27}{8}$

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35 [Intermediate] • Target: The Taylor series formula

Use the Maclaurin series for e^u (with $u = 2x^2$) to find the coefficient of x^8 in the Maclaurin series for e^{2x^2}.

a. $\frac{2}{3}$

b. $\frac{1}{2520}$

c. $\frac{5}{3}$

d. $\frac{32}{3}$

36 [Intermediate] • Target: Common Maclaurin series

Using the Lagrange Error Bound $|R_n(x)| \leq \frac{M}{(n+1)!}|x-a|^{n+1}$ with $M = 6$, $n = 2$, and $|x-a| = 1$, find the error bound.

a. 2

b. $\frac{7}{6}$

c. 3

d. 1

37 [Intermediate] • Target: Using known series to build new series

Using the Lagrange Error Bound $|R_n(x)| \leq \frac{M}{(n+1)!}|x-a|^{n+1}$ with $M = 5$, $n = 3$, and $|x-a| = 1$, find the error bound.

a. $\frac{5}{6}$

b. $\frac{5}{24}$

c. $\frac{29}{24}$

d. $\frac{1}{4}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The Lagrange error bound • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Using the Lagrange Error Bound $|R_n(x)| \leq \frac{M}{(n+1)!}|x-a|^{n+1}$ with $M = 4$, $n = 2$, and $|x-a| = 1$, find the error bound.

a. $\frac{5}{3}$

b. 2

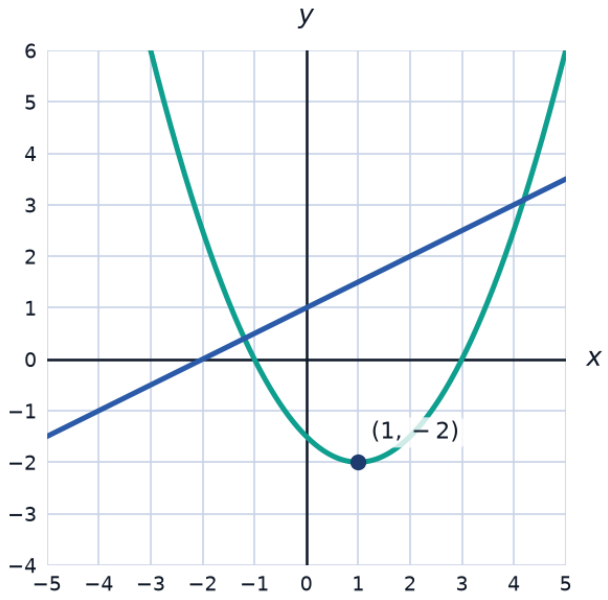
c. $\frac{5}{6}$

d. $\frac{2}{3}$

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39 **[Challenging]** • **Target: Radius of convergence of Taylor series** • **Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests**

Using $\cosh(8x) = \frac{e^{8x} + e^{-8x}}{2} = \sum_{n=0}^{\infty} \frac{(8x)^{2n}}{(2n)!}$, find the coefficient of x^6.



a. $\frac{32768}{15}$

b. $\frac{16384}{315}$

c. $\frac{16384}{45}$

d. $\frac{16429}{45}$

40 **[Challenging]** • **Target: Visualizing Taylor polynomial approximations** • **Cross-Lesson: 6.01 Power Series and Interval of Convergence**

Using $\cosh(4x) = \frac{e^{4x} + e^{-4x}}{2} = \sum_{n=0}^{\infty} \frac{(4x)^{2n}}{(2n)!}$, find the coefficient of x^4.

a. $\frac{35}{3}$

b. $\frac{128}{3}$

c. $\frac{32}{3}$

d. $\frac{32}{15}$

41 **[Challenging]** • **Target: Common Maclaurin Series ($e^x, \sin x, \cos x$)** • **Cross-Lesson: 1.01 Integration by Parts**

What is the coefficient of x^4 in the Maclaurin series for e^{-2x}?

a. $\frac{2}{3}$

b. $\frac{16}{24} = \frac{2}{3}$

c. $-\frac{2}{3}$

d. $\frac{16}{4}$

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42 [Challenging] • Target: Constructing Taylor Polynomials from Derivatives • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

Which formula gives the n-th degree Taylor polynomial $T_n(x)$ centered at $x = c$?

a. $\sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x-c)^k$

b. $\sum_{k=0}^{n} f^{(k)}(c)(x-c)^k$

c. $\sum_{k=0}^{n} \frac{f(c)}{k!}(x-c)^k$

d. $\frac{f^{(n)}(c)}{n!}(x-c)^n$

43 [Challenging] • Target: Constructing Taylor Polynomials from Derivatives • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

If a function f has $f(2) = 5$, $f'(2) = -3$, $f''(2) = 8$, what is $T_2(2.1)$?

a. 4.74

b. 4.70

c. 4.82

d. 5.04

44 [Challenging] • Target: Manipulating Known Taylor Series • Cross-Lesson: 3.01 Parametric Equations

Find the Maclaurin series for $\frac{\sin x - x}{x^3}$.

a. $-\frac{1}{6} + \frac{x^2}{120} - \ldots$

b. $\frac{1}{6} - \frac{x^2}{120} + \ldots$

c. $1 - \frac{x^2}{6} + \ldots$

d. $-\frac{x^3}{6} + \ldots$

45 [Challenging] • Target: Lagrange Error Bound (Taylor's Remainder Theorem) • Cross-Lesson: 3.02 Vector-Valued Functions

What is the Lagrange Error Bound formula for $|R_n(x)| = |f(x) - P_n(x)|$ centered at c?

a. $|R_n(x)| \le \frac{M}{(n+1)!}|x-c|^{n+1}$, where $M = \max|f^{(n+1)}(t)|$

b. $|R_n(x)| \le \frac{M}{n!}|x-c|^n$

c. $|R_n(x)| \le |f^{(n+1)}(c)|$

d. $|R_n(x)| \le \frac{1}{(n+1)!}$

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